Three Number Sum

AlgoExpert · Medium · Arrays

The problem

- All triplets of distinct integers summing to target, each triplet ascending and the list sorted.
- Sort then sweep with two pointers.

Intuition

- Fix one number, then find pairs in the sorted remainder with two pointers.

Approach

- 1. Sort the array
- 2. For each index, set left and right pointers after it
- 3. Move them based on the current sum
- 4. Record exact matches

Java solution

```java
public class ThreeNumberSum {
    public static List<Integer[]> threeNumberSum(int[] array, int target) {
        Arrays.sort(array);
        List<Integer[]> out = new ArrayList<>();
        for (int i = 0; i < array.length - 2; i++) {
            int l = i + 1, r = array.length - 1;
            while (l < r) {
                int s = array[i] + array[l] + array[r];
                if (s == target) { out.add(new Integer[]{array[i], array[l], array[r]}); l++; r--; }
                else if (s < target) l++; else r--;
            }
        }
        return out;
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(n)

Recap

- Fix one number, then find pairs in the sorted remainder with two pointers.
- Complexity: O(n^2)